TN SKILL PROJECT | REQUIREMENT ANALYSIS PHASE

Auto Ticket Classification using Flow Designer

Customer Journey Map & Requirement Analysis

1. OBJECTIVE OF THE CUSTOMER JOURNEY

The customer journey maps every step that a support ticket goes through, from the moment an end user faces an issue until the ticket is resolved and closed. Mapping the journey exposes where delays, errors and frustration occur during manual triage, and it shows which requirements Flow Designer must satisfy to automate classification and routing.

2. ACTORS INVOLVED

Actor	Role	Goal in the journey
Karthik	Tier-1 IT Support Engineer	Stop spending hours on manual triage; focus on real troubleshooting.
End User / Customer	Employee raising an incident or request	Get the issue acknowledged, assigned and resolved quickly.
Tier-2/3 Engineer	Network, Database, Security, Identity teams	Receive only tickets that truly belong to their group.
IT Manager	Service Desk owner	Keep the unassigned queue at zero and avoid SLA breaches.

3. CURRENT (AS-IS) CUSTOMER JOURNEY

The table below traces the manual process. Emotions rise and fall with queue delay, and the sharpest pain sits at Manual Triage and Routing.

Stage	Customer / Agent Actions	Touchpoint	Emotion	Pain Points
1. Awareness	User faces an issue (VPN down, password locked) and decides to raise a ticket.	Email / Portal	Worried	Unsure which category to choose; writes vague text like "Internet down".
2. Ticket Creation	User submits incident; ticket lands in the central unassigned queue.	Service Portal	Hopeful	No instant ownership or acknowledgement of who is handling it.
3. Manual Triage	Karthik reads every description, picks category, priority and group (2–5 min each).	Service Desk dashboard	Overwhelmed	100+ tickets/day; urgent outages trapped in the unassigned pool.
4. Routing	Ticket assigned to a resolver group; wrong assignment bounces back.	Assignment group queue	Anxious	~20–30% re-routing ("ping-pong"); risk of SLA breach.
5. Resolution	Tier-2/3 engineer analyses and fixes the issue.	Incident record	Neutral	Delayed first response increases MTTR.

Stage	Customer / Agent Actions	Touchpoint	Emotion	Pain Points
6. Closure & Feedback	User is notified and closes the ticket; gives feedback.	Email / Portal survey	Frustrated	Low CSAT because of long queue wait times.

4. FUTURE (TO-BE) JOURNEY WITH FLOW DESIGNER

Once classification is automated at ticket creation, the manual triage stage is removed and the journey becomes continuous and predictable.

Stage	Automated Experience with Flow Designer	Emotion	Improvement
1. Awareness	User logs the issue using guided fields and a short description.	Confident	Structured input improves keyword accuracy.
2. Ticket Creation	Trigger fires the moment a new Incident/Request is created.	Reassured	Instant acknowledgement (< 10 seconds).
3. Auto Classification	Flow parses short description and body for keywords (VPN, Password Reset, Hardware Failure) and sets Category, Subcategory, Priority.	Relieved	Zero manual triage; consistent rules for identical issues.
4. Auto Routing	Ticket routed to the correct queue (Network, Security, Identity) with SLA timer and alert.	Calm	Routing accuracy > 90–95%; bounce rate < 5%.
5. Resolution	Specialist receives a correctly tagged ticket and starts work at once.	Focused	Lower MTTR; Tier-1 time spent on resolution.
6. Closure & Feedback	User is notified in real time and closes with feedback.	Satisfied	Higher CSAT and SLA compliance.

5. KEY MOMENTS OF TRUTH

- **First acknowledgement:** the user must see the ticket owned within seconds, not hours.
- **Correct first assignment:** a ticket that reaches the right team the first time avoids ping-pong and protects the SLA.
- **Priority accuracy:** critical outages must never wait in the unassigned pool.

6. REQUIREMENTS DERIVED FROM THE JOURNEY

FUNCTIONAL REQUIREMENTS	NON-FUNCTIONAL REQUIREMENTS
• FR1: Trigger on creation of every new Incident or Request item. • FR2: Parse short description and detailed body text for keywords. • FR3: Match terms against configurable rules or decision tables. • FR4: Auto-set Category, Subcategory, Urgency and Priority. • FR5: Auto-assign to the correct resolver group.	• NFR1: Classification completes in under 10 seconds. • NFR2: Accuracy of at least 90% on standard incident datasets. • NFR3: Rules are low-code and easy for admins to update. • NFR4: Reliable at peak load with no queue build-up. • NFR5: Every automated action is logged for audit.

• FR6: Notify the assigned group and attach SLA timers. • FR7: Send unmatched tickets to a default review queue.	• NFR6: Works within the platform using standard APIs.

7. SUCCESS METRICS & EXPECTED OUTCOME

Success Metric	Current (Manual)	Target (Automated)
Triage time per ticket	3 – 5 minutes	< 1 second
Routing / re-assignment	20–30% re-routed	< 5% misallocation
SLA breaches (response)	Frequent at peak load	Reduced by over 40%
Agent time on sorting	40% of shift	Near zero

8. CONCLUSION

The journey analysis shows that the biggest breakdown occurs at the manual triage and routing stages, not at technical resolution. Automating these stages with Flow Designer removes queue delay, improves routing accuracy, and moves the customer experience from frustration and waiting to fast, confident and satisfied. The requirements listed above form the basis for the design and build phases that follow.